Receipt OCR and Item Details

Part	How it works
Image input	Choose photo, device camera/gallery, or live webcam capture.
OCR engine	Tesseract.js creates an English OCR worker and reads text from the selected image.
Parser	receiptParser.ts detects merchant names, total/payment lines, rupee/dollar amounts, and line items.
Auto-filled fields	Description, total amount, optional notes, item count, confidence, and warnings.
Notes output	Scanned item names, quantities when detected, individual item amounts, and receipt total.
Verification	The UI warns the user to verify OCR results before saving because real receipts can be noisy.

AI type used

The current project does not call a cloud LLM. The 'AI' behavior is local deterministic logic: keyword scoring, confidence calculation, practical advice rules, and OCR-assisted receipt parsing.

7. Analytics, Charts, and Animations

- StatsOverview shows Total Spent, This Month, and AI Categorized. Each card opens a modal with matching expense details.
- SpendingChart shows the top category totals as progress bars.
- CategoryPieChart renders a custom SVG donut/pie chart with stable category colors.
- The full chart modal supports Day, Last 24h, Month, and Year filters plus category/description search.
- Chart animations in index.css include modal fade-in, panel scale-in, pie pop-in, slice drawing, and center-text fade.
- Reduced-motion support disables chart animations for users who prefer less motion.

8. macOS Desktop App Packaging

How the app became a Mac app

The project was not converted into Electron. Instead, the Vite build was copied into a normal macOS .app bundle. The bundle starts a tiny local server and opens the app URL in the browser, so the user can launch ExpenseAI by double-clicking the desktop app.

Piece	Purpose
ExpenseAI.app/Contents/Info.plist	Defines bundle name, executable, identifier, version, package type, and LSUIElement.
ExpenseAI.app/Contents/MacOS/ExpenseAI	zsh launcher that checks localhost, writes/starts a LaunchAgent, waits for the server, and opens the app URL.
ExpenseAI.app/Contents/Resources/site	Contains the built Vite site: index.html plus hashed JS/CSS assets.
ExpenseAI.app/Contents/Resources/server.mjs	Node fallback static server if Python 3 is unavailable.